1. Project Description

CliniScan is an AI-based medical imaging system designed to automatically analyse chest X-ray images and detect abnormalities using deep learning techniques. Chest X-rays are widely used in diagnosing lung diseases such as pneumonia, pulmonary fibrosis, and pleural effusion.

The goal of this project is to build an intelligent system capable of automatically identifying abnormalities in chest X-ray images. The system combines two deep learning approaches:

1. Multi-Label Classification – determines which diseases are present in the X-ray.

2. Object Detection – identifies the location of abnormalities using bounding boxes.

By combining classification and detection, the system provides both diagnostic prediction and spatial localization of disease regions.

2. Dataset Used

The dataset consists of 5000 chest X-ray images with corresponding annotations describing abnormalities present in each image.

Each annotation includes:

• image_id
• class_name
• class_id
• x_min, y_min
• x_max, y_max

These coordinates define bounding boxes that indicate the location of abnormalities within the X-ray image.

The dataset contains 15 pathology classes plus a background class for detection, giving a total of 16 classes.

1. Classification Model –

1. Overview

This module implements a multi-label chest X-ray classification system using deep learning.
The model predicts multiple abnormalities in a single image and includes:

- Data augmentation
- Class imbalance handling
- Hyperparameter tuning
- Threshold optimization
- Performance evaluation (F1, AUC)
- Grad-CAM visualization

2. Data Preprocessing and Augmentation

To improve generalization, Albumentations is used.

Training Augmentations:

- Resize (224×224)
- Horizontal Flip
- Rotation (±10°)
- Brightness & Contrast adjustment
- CLAHE (enhances X-ray contrast)
- Gaussian Noise
- Random Resized Crop
- Normalization

Validation:

- Only Resize + Normalize

👉 These are medically safe augmentations that simulate real-world X-ray variations.

3. Dataset and DataLoader

- Custom dataset: ChestXrayDataset
- Input: CSV + images
- Output: image tensor + multi-label vector

DataLoader:

- Batch size: 16
- Shuffle: True (train), False (val)

4. Model Architecture

- Pretrained deep learning model (Transfer Learning)
- Modified final layer → 15 output classes

Transfer learning helps extract medical features efficiently.

5. Handling Class Imbalance

Medical datasets have rare disease classes.

Solution:

- Weighted BCEWithLogitsLoss
- Rare classes → higher weight
- Frequent classes → lower weight

6. Loss Function and Optimizer

- Loss: BCEWithLogitsLoss
- Optimizer: AdamW
  - LR = 3e-4
  - Weight decay = 1e-4

7. Training Process

Each epoch consists of:

Training:

- Forward pass
- Loss calculation
- Backpropagation
- Weight update

Validation:

- Predictions using sigmoid
- Collect probabilities
- Compute metrics

8. Threshold Optimization

Instead of fixed 0.5 threshold:

- Try thresholds from 0.1 → 0.9
- Select threshold giving best F1 score

9. Evaluation Metrics

- Macro F1 Score → handles imbalance
- AUC Score → measures prediction quality

10. Model Saving

- Best model saved based on F1 score
- Final model also saved

11. Grad-CAM Visualization

Used for model interpretability.

Process:

- Extract feature maps
- Compute gradients
- Generate heatmap
- Overlay on image

Output:

- Heatmap showing important regions in lungs

12. Performance Visualization

Graphs generated:

- Loss vs Epoch
- F1 Score vs Epoch
- AUC vs Epoch
- Threshold vs F1

13. Key Improvements

- Data augmentation improved generalization
- Weighted loss handled imbalance
- Threshold tuning improved F1
- Grad-CAM improved interpretability

2. Detection Model –

1. Overview

This module performs object detection on chest X-ray images to locate abnormalities using bounding boxes.

The system is based on:

- Faster R-CNN architecture
- Transfer learning
- Custom medical-safe augmentations

2. Dataset and Preprocessing

Custom dataset: ChestXrayDetectionDataset

Features:

- Reads bounding boxes from CSV
- Removes invalid boxes
- Converts images to tensor
- Caches images for faster training

Output:

- Image tensor
- Target dictionary:
  - boxes
  - labels
  - area
  - iscrowd

3. Data Augmentation (Medical Safe)

Custom augmentation class: MedicalAugment

Applied only on training data:

- Horizontal Flip
- Small Rotation (±10°)
- Scaling (±10%)
- Brightness adjustment
- Contrast adjustment
- Mild Gaussian noise

Model Architecture

- Faster R-CNN with MobileNet backbone
- Pretrained weights used

- Fine-tuned for 16 classes

Enables both classification + localization.

5. Training Configuration

- Optimizer: AdamW
- Learning rate: 1e-4
- Scheduler: StepLR
- Batch size: 4

6. Training Loop

For each epoch:

Training:

- Forward pass → returns loss components:
  - Classification loss
  - Bounding box regression loss
  - Objectness loss
  - RPN loss
- Total loss = sum of all losses
- Backpropagation + update

Validation Process

Loss Calculation:

- Model in train mode (required for Faster R-CNN loss)

Prediction Evaluation:

- Model in eval mode
- Generate predictions

Evaluation Metrics

Computed:

- Precision
- Recall
- F1 Score

Logic:

- IoU ≥ 0.5 → correct detection
- Else → false positive

IoU (Intersection over Union)

Used to compare predicted and ground truth boxes:

- Measures overlap between boxes
- Threshold = 0.5

Visualization of Predictions

Function: visualize_predictions

- Predicted boxes → RED
- Ground truth → GREEN

Model Saving

- Best model saved based on validation loss
- Checkpoints saved every epoch

Performance Tracking

- Train loss
- Validation loss
- Precision / Recall / F1

Graphs generated:

- Loss curve